GREENNIRMAN

India's Green Building Assessment Platform

Empowering Sustainable Construction Across 758 Indian Districts

758	72	21 Yrs
Districts	Materials	NASA Climate Data

March 2026 | ECBC 2017 · NBC 2016 · GRIHA · NASA POWER | Priyanshu Kumar · GITM

THE CHALLENGE

India's Built Environment Crisis

40%

of national energy consumed by buildings

22%

of India's total CO_2 emissions

9.2M+

9.2M new buildings annually (11.3M by 2028)

~9x

cooling energy demand growth by 2050

Most green building tools work only in metros. GreenNirman brings district-level precision to every corner of India.

WHAT IS GREENNIRMAN?

A browser-based green building assessment platform built exclusively for India — covering all 758 districts across 36 states and UTs.

Generates a professional PDF assessment report that architects, engineers, and homeowners can submit for green building approvals — fully offline, no server needed.

758 Districts

All states & UTs covered

GRIHA + ECBC

Dual compliance check

NASA Climate

21-yr real weather data

Energy Savings

kWh + Rs./year calculated

72 Materials

Carbon rated A+ to E

Water Risk

CGWB district data

Standards: ECBC 2017 · NBC 2016 · IS 1893:2016 · GRIHA v2019 · Eco Niwas Samhita 2021

GEOGRAPHIC COVERAGE

758
Total Districts
All 36 states & UTs

670
With NASA Climate
Real 21-yr weather data

758
With Insights
Seismic · Flood · Soil · Water

5 ECBC Climate Zones

Zone 1 — Hot & Dry	Jaipur, Jodhpur, Jaisalmer, Nagpur	**~180**
Zone 2 — Warm & Humid	Mumbai, Chennai, Kochi, Visakhapatnam	**~140**
Zone 3 — Composite	Delhi, Bengaluru, Pune, Lucknow	**~250**
Zone 4 — Temperate	Mysuru, Coimbatore, Shillong, Darjeeling	**~80**
Zone 5 — Cold	Leh, Srinagar, Shimla, Dehradun	**~108**

NASA POWER CLIMATE DATA

Temperature

T2M

Monthly mean air temp at 2m height

Powers OTTV/RTTV energy calculations

Humidity

RH2M

Monthly mean relative humidity at 2m

Moisture risk + ventilation design

Solar Radiation

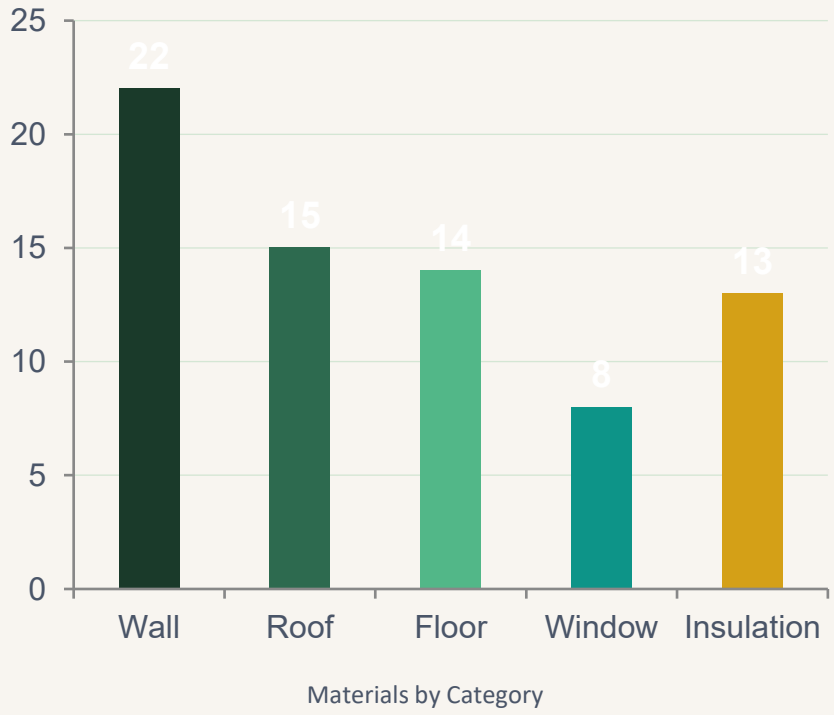
ALLSKY_SFC_SW_DWN

All-sky surface shortwave irradiance

Solar heat gain + PV yield estimation

21-year averages (2000–2020) fetched via NASA POWER API for each district's latitude/longitude · 670 districts with real data · 88 use zone average

72 MATERIALS DATABASE



Carbon Rating Scale

A+ — < 0.05 kg CO_2/kg
Adobe, Stone, Laterite, CSEB, Thatch

A — 0.05–0.15
Fly Ash Brick, Bamboo, Kota Stone

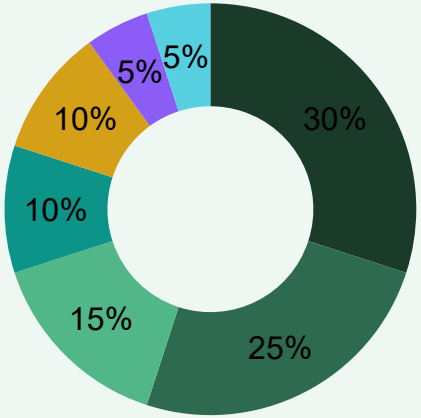
B — 0.15–0.25
Hollow Block, Rat-Trap, Ferro-cement

C — 0.25–0.50
Burnt Clay Brick, AAC Block, RCC Slab

D — 0.50–2.00
All glazing types, Rock Wool

E — > 2.00
GI Sheet, EPS/XPS Polystyrene

7-PARAMETER SCORING METHODOLOGY



- Embodied Carbon 30%
- Thermal 25%
- Availability 15%
- Cost 10%
- Durability 10%
- Renewable 5%
- Water 5%

30% **Embodied Carbon**
CO_2 in material manufacturing + transport

25% **Thermal Performance**
U-values vs ECBC limits, OTTV/RTTV

15% **Material Availability**
Local sourcing — reduces transport carbon

10% **Cost Efficiency**
User cost vs CPWD DSR 2024 baseline

10% **Durability**
Average material service life

5% **Renewable/Recycled**
% renewable or recycled materials

5% **Water Management**
RWH adoption + CGWB scarcity risk

GRIHA RATING + ECBC COMPLIANCE

GRIHA Star Rating (points out of 24)

< 8 pts	★	Below baseline
8–11 pts	★★	ECBC minimum
12–15 pts	★★★	20-30% better
16–19 pts	★★★★	30-50% better
20–24 pts	★★★★★	Net-zero direction

ECBC 2017 U-Value Limits (W/m²K)

Zone	Wall	Roof	Window
Zone 1 Hot & Dry	0.44	0.33	3.30
Zone 2 Warm & Humid	0.44	0.33	3.30
Zone 3 Composite	0.44	0.33	3.30
Zone 4 Temperate	0.44	0.33	3.30
Zone 5 Cold	0.30	0.20	2.00

DISTRICT INSIGHTS — LOCAL RISK DATA

Seismic Zone

BIS IS 1893:2016 earthquake risk (I–V)

Flood Risk

Low / Moderate / High / Very High

Soil Type

Sandy / Black cotton / Laterite / Rocky

Water Scarcity

CGWB groundwater stress level

Construction Cost

Rs./sqft range from CPWD DSR 2024

Best Wall Materials

Top 3 climate-optimised wall choices

Best Roof Materials

Top 3 climate-optimised roof choices

Climate Advice

District-specific passive design guidance

CONTRIBUTION TO INDIA'S

Sustainability Goals

Net Zero 2070

Quantifies operational carbon per building — drives material choices toward low carbon

NDC 2030 Target

Embodied + operational scoring pushes 45% emissions intensity reduction goal

PMAY Housing

Affordable tier ensures EWS/LIG housing gets green building guidance

Jal Jeevan Mission

CGWB water data + RWH sizing in 180+ water-critical districts

Smart Cities

District-level data supports city green building planning + bylaws

ECBC Enforcement

Makes compliance accessible beyond large commercial projects

Building India's Green Future — One District at a Time

PROFESSIONAL PDF ASSESSMENT REPORT

Generated entirely in the browser — no server, no data upload. Submit to authorities, banks, and government schemes.

1 Cover Page
Project, district, zone, area, building type, date

2 Executive Summary
Score, GRIHA stars, ECBC pass/fail at a glance

3 Score Breakdown
7 parameters with radar chart and explanations

4 ECBC Compliance
Wall/Roof/Window vs limits, OTTV/RTTV/RETV

5 Energy Analysis
Annual kWh savings + Rs./yr + 25-year lifecycle cost

6 Carbon Footprint
Embodied + operational CO_2, tree equivalent

7 Material Summary
All 5 materials — cost, carbon, U-value, life, rating

8 District Insights
Seismic, flood, soil, water, construction cost

9 Recommendations
Top prioritised actions with expected impact

10 Methodology
Standards, data sources, calculation basis

GreenNirman

Building India's Green Future · One District at a Time

758 Districts	**72 Materials**	**670 NASA**	**7 Parameters**	**5 Zones**

🔒 This app does not collect, store, or transmit any user data. All calculations happen locally in your browser.

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